ADRIAN EGEA | SENIOR SOFTWARE ENGINEER

C++ | Systems Architecture | Real-Time Systems | Unreal Engine 5 | Graphics | Vulkan | OpenGL

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PROFESSIONAL SUMMARY

Software Engineer with **9+ years of experience** specializing in software architecture, gameplay and systems programming, performance optimization, cross-platform development, and Modern C++ (17/20/23). Graphics API enthusiast with a custom **Vulkan 3D Game Engine**. I've taken part in multiple **AA/AAA Unreal Engine** projects across all production stages (pre-release, mid-production, and greenfield), such as **Lords of the Fallen** (2023, PS5, XSX, PC), **Gothic Remake** (2026, PS5, XSX, PC), and **Remothered Tormented Fathers Remaster** / **Remothered Broken Porcelain Remaster** (2026, PS5, XSX, Switch 2, PC). I bring a strong engineering mindset, adaptability, and a hands-on approach to solving complex technical challenges.

TECHNICAL SKILLS

- **Programming Languages**: C++ (17/20/23), C, Python, JavaScript, HTML, CSS
- **Software Engineering**: Software Architecture, Object-Oriented, Data-Driven, TDD, Pair-Programming
- **Graphics Programming**: Vulkan, OpenGL, WebGL, GLSL, 3D, 2D, Maths, GPU optimization, PBR
- **Build and Automation**: Linux, CMake, CI/CD, Jenkins, TeamCity, Git, Gitea, Docker, Portainer
- **Frameworks and Tools**: Unreal Engine 5, GAS, Unreal Build Tool, PlayStation SDK, Switch SDK, XBox SDK
- **Debug and Profiling**: Vulkan Validation Layers, RenderDoc, Nvidia Nsight, Tracy, ImGUI, Unreal Profiler

PROFESSIONAL EXPERIENCE

Mobile C++ Programmer | King Aug 2026 – Present (3m)

- Contribute to the implementation and maintenance of C++ production software in a mobile development environment.

Principal UE5 C++ Programmer | Radical Dreams Entertainment Feb 2026 – Aug 2026 (7m)

- Developed cross-platform **Remaster** ports for **Remothered Tormented Fathers** and **Remothered Broken Porcelain** (PC, PS5, XSX, Switch 2).
- Contributed to **porting** existing software to Unreal Engine 5 and resolving platform-specific issues.
- Implemented and maintained gameplay and system-level functionality, UI components, and cinematic systems.
- Set up **CI/CD** infrastructure using **TeamCity** to support automated builds and development workflows.

Unreal Engine 5 C++ Programmer | GRIP Studios Apr 2025 – Jan 2026 (10m)

- Designed and implemented a **Modding system** supporting PC and console platforms for unannounced **Simulation** project (PC, PS5, XSX).
- Developed gameplay systems and refactored existing **Mutable** architecture to support runtime generation of characters and vehicles.
- Built debugging and visualization tools to improve development workflows and facilitate issue diagnosis.

Unreal Engine 5 C++ Programmer | Underdog Studio (CI Games) Mar 2024 – Oct 2024 (8m)

- Developed a Quest management system for an unannounced multiplayer **Survival** project (PS5, XSX, PC) and Quest system state replication to support cooperative multiplayer gameplay.

Unreal Engine 5 C++ Programmer | HexWorks (CI Games) Aug 2021 – Mar 2024 (2y 8m)

- Designed, implemented, and maintained core software systems for **Lords of the Fallen** (2023, PS5, XSX, PC), including quest management, save-game persistence, and data serialization.
- Developed gameplay systems involving inventory management, character statistics, and audio integration.
- Contributed to multiplayer networking, state replication, and gameplay ability systems (GAS).
- Investigated and resolved critical software defects during development and post-release maintenance, including patches and DLC releases.
- Improved asynchronous assets and data loading workflows to optimize runtime performance.

Unreal Engine 4 C++ Programmer | Alkimia Interactive (THQ Nordic) May 2020 – Jul 2021 (1 yr 3m)

- Worked on **Gothic Remake** (2026; PS5, XSX, PC).
- Developed character and creature movement systems and animation-related functionality for quadrupeds.
- Migrated existing ALS Plugin implementation to C++ to improve code performance and maintainability.
- Implemented and debugged AI functionality for small creature behaviors (birds, rats, fishes…).
- Contributed to movement systems and gameplay abilities within an established software architecture.

Lead Unreal Engine 4 C++ Programmer | Limitless Apr 2018 – Mar 2020 (2y)

- Worked on a cancelled, unannounced **MOBA** project (PC).
- Led a five-developer programming team, coordinating implementation tasks and technical decisions.
- Defined software architecture and implemented networking functionality for multiplayer.
- Developed automated build pipelines using Jenkins and Git.
- Created internal development tools and Git automation scripts to improve workflows for the art team.

Graphics Developer | Cl3ver May 2017 – Apr 2018 (1 yr)

- Developed C++ and OpenGL functionality for an **Oculus Rift** application.
- Implemented 3D data importer plugins for **Rhino** and **SketchUp**.

Full-Stack / Android Developer | Tretanto Feb 2015 – Apr 2016 (1 yr 3m)

- Developed Web, Android applications and social networking features using JS, jQuery, AngularJS, Android, PHP, Symfony2, Doctrine, and MySQL.

SIDE PROJECTS

C++ / OpenGL / Vulkan 3D Game Engine Nov 2017 – Present (9y)

- Custom Vulkan 3D engine built for graphics and systems learning and research. GitHub Repo
- **Tech**: C++, CMake, Python, Vulkan, OpenGL 4.x, GLSL, SPIRV
- **Features**: 2D, 3D, Hybrid ECS, Automatic Batching, Instanced Rendering, Data Driven Shaders Builder, Texture Atlases, Animation Spritesheets, Built-In Custom UI, Scene management, Perspective / Orthographic Camera, Stencil, Basic Input, Events Manager, Math module, Loading 3D models with GLTF, Skeletal Meshes, Skeletal Animation, GPU Skinning, Render Graph, Render Passes, PBR Shading, Directional Light, Shadow Mapping, Materials system.

JavaScript + WebGL 2D Game Engine Jul 2013 – May 2017 (3y 11m)

- Lightweight Web 2D engine. GitHub Repo
- **Tech**: JavaScript, WebGL, GLSL
- **Features**: 2D, Texture Atlases, Animation Spritesheets, Scene management, Orthographic Camera, Basic Input

Game AI Project | Pathfinding A*, Flocking, State Machines Apr 2016 – Jul 2016 (4m)

- Implementation of autonomous agent behaviors and navigation algorithms. GitHub Repo
- **Tech**: Java, XML
- **Features**: Tiled Map, Pathfinding A*, Flocking, State Machines, Steering Behaviours

Transpiler - Custom C++-Like Language Oct 2022 - Present

- Simplified C++ to modern C++ transpiler. GitHub Repo
- **Tech**: C++, CMake
- **Features**: Lexer, Parser, AST, Semantic Analysis, Code Generation

rss-to-md and md-to-html Python tools Jun 2025

- **rss-to-md**: Download RSS info in Markdown format. GitHub Repo
- **md-to-html**: Convert Markdown to HTML. GitHub Repo
- **Tech**: Python, HTML, CSS, JSON, XML
- **Features**: RSS download, RSS Cache, Human-like behaviour to avoid bans, categories, sidebar navigation tree, sidebar table of content, pretty link cards, search bar

EDUCATION

- B.Tech in Computer Science and Engineering (University of Murcia - FIUM, 2012 - 2017)